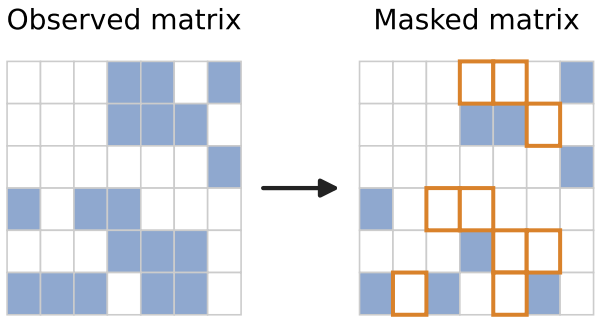
Controlled masking

PBMC 3K (human) + Paul15 (mouse)



genes × cells

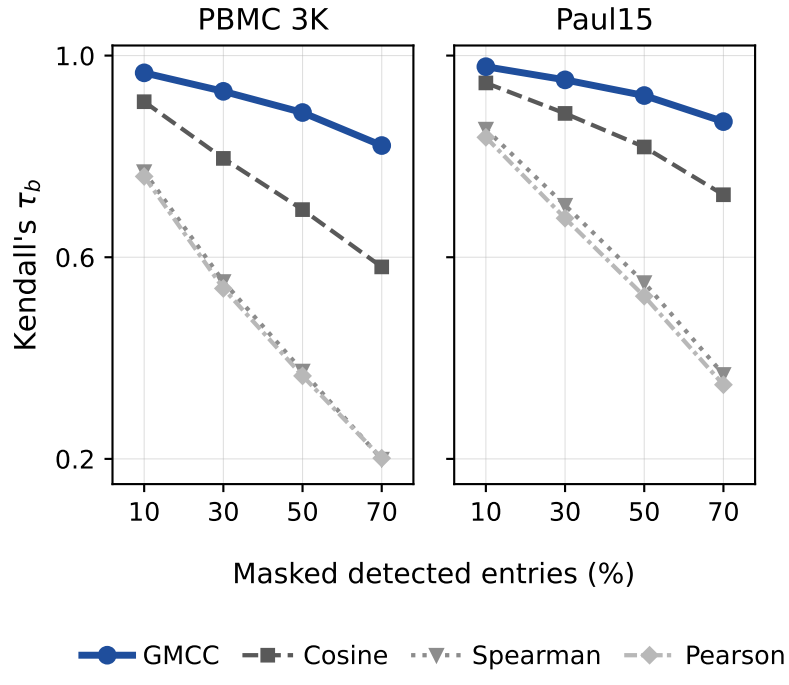
Detected entries randomly masked to zero (10–70%)

20 independent realisations per condition

GMCC | Pearson | Cosine | Spearman

Gene-pair rank robustness

Higher is more robust

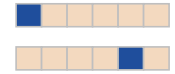


GMCC also showed low magnitude change at moderate–severe masking

Main findings

Highest rank robustness

GMCC in both datasets, with μ recomputed from each masked matrix



Shared zeros inflate GMCC similarity

rarely detected genes look alike because both are mostly zero

Advantage holds after detection filter

genes detected in ≥10% of cells: GMCC still the most robust

GMCC and Cosine are complementary

GMCC → sparsely detected genes
Cosine → highly detected gene pairs

Robustness ≠ biological accuracy